

Jonas Conneryd

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Education

Lund University Ph.D. in Theoretical Computer Science Advisor: Prof. Tatyana Turova Lic. Eng. in Computer Science Thesis: <i>On the Average-Case Proof Complexity of Graph Coloring</i> Opponent: Prof. Nutan Limaye, ITU Copenhagen	Lund, Sweden 2026 (expected) Thesis defended May 2025
KTH Royal Institute of Technology M.Sc. in Mathematics (joint with Stockholm University) Thesis: <i>Geometric Bounds for Steklov Eigenvalues on Graphs</i> (awarded Mittag-Leffler Prize) B.Sc. in Engineering Physics Thesis: <i>Explicit Symplectic Integrators for Non-Separable Hamiltonians in Molecular Dynamics</i>	Stockholm, Sweden June 2021 June 2019

Research Interests

Computational complexity theory, particularly proof complexity with an algebraic flavor and the complexity of (random) constraint satisfaction problems.

Publications

- [1] Jonas Conneryd, Yassine Ghannane, and Shuo Pang. *Lower Bounds for CSP Hierarchies Through Ideal Reduction*. SODA 2026.
- [2] Jonas Conneryd, Kilian Risse, and Dmitry Sokolov. *Graph Coloring Is Hard on Average for Polynomial Calculus over Roots of Unity*. To be incorporated into a larger project also involving Yassine Ghannane, Jakob Nordström, and Shuo Pang. 2025.
- [3] Jonas Conneryd, Susanna F. de Rezende, Jakob Nordström, Shuo Pang, and Kilian Risse. *Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz*. FOCS 2023.
- [4] Jonas Conneryd. *Geometric Bounds for Steklov Eigenvalues on Graphs*. M.Sc. thesis. Stockholm University, 2021.

Talks

ACM-SIAM Symposium on Discrete Algorithms (SODA) <i>Lower Bounds for CSP Hierarchies Through Ideal Reduction</i>	Vancouver, BC, Canada Winter 2026
KTH Royal Institute of Technology <i>Lower Bounds for CSP Hierarchies Through Ideal Reduction</i>	Stockholm, Sweden Fall 2025
Oxford Proof Complexity Workshop <i>Lower Bounds for CSP Hierarchies Through Ideal Reduction</i>	Oxford, United Kingdom Summer 2025
Proof Complexity and Beyond, Oberwolfach Workshop 2413 <i>Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz</i>	Oberwolfach, Germany Spring 2024
Algorithmic Research Cooperation around Øresund (ARCO) <i>Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz</i>	Odense, Denmark Fall 2023
IEEE Symposium on Foundations of Computer Science (FOCS) <i>Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz</i>	Santa Cruz, CA, USA Fall 2023

Research Visits and Workshops

Swiss Winter School on Theoretical Computer Science Visiting researcher for a winter school jointly organized by EPFL and ETH Zürich.	Zinal, Switzerland Winter 2026
KTH Royal Institute of Technology Month-long visit to the theory group.	Stockholm, Sweden Fall 2025
University of Oxford Visiting researcher for the workshop <i>Proof Complexity</i> in 2024 and 2025 .	Oxford, United Kingdom Summers 2024, 2025
Centre International de Rencontres Mathématiques Visiting researcher for the <i>Complexity as a Kaleidoscope</i> research school.	Marseille, France Spring 2025
DIMACS, Rutgers University Visiting researcher for the <i>Frontiers in Complexity Workshop</i> .	New Brunswick, NJ, USA Summer 2024
McGill University Week-long visit to the group of Prof. Robert Robere.	Montréal, Canada Summer 2024

Mathematisches Forschungsinstitut Oberwolfach

Visiting researcher for the workshop *Proof Complexity and Beyond*.

Oberwolfach, Germany

Spring 2024

Chalmers University of Technology

Visiting researcher for the *Workshop on Algebra and Computation*.

Gothenburg, Sweden

Summer 2023

Simons Institute for the Theory of Computing at UC Berkeley

2-month visiting graduate student for the program *Satisfiability: Extended Reunion*.

Berkeley, CA, USA

Spring 2023

Schloss Dagstuhl

Visiting researcher for the workshop *Theory and Practice of SAT and Combinatorial Solving*.

Dagstuhl, Germany

Fall 2022

Honors and Scholarships

- WASP travel grant for one-month visit to McGill University; 2026
- Oberwolfach Leibniz Graduate Student; Mathematisches Forschungsinstitut Oberwolfach, 2024
- WASP travel grant for two-month visit to Simons Institute for the Theory of Computing at UC Berkeley; 2023
- Accepted as *WASP* affiliated student; 2021
- Mittag-Leffler Prize for outstanding M.Sc. theses in mathematics; Stockholm University, 2021
- Ingenjör Ernst Johnson Scholarship for outstanding academic achievements; KTH, 2020, 2021
- Henrik Göransson Sandviken Scholarship for outstanding academic achievements; KTH, 2018
- University Merit Scholarship for outstanding academic achievements; KTH, 2018, 2019, 2020, 2021

Service

Reviewed papers for the following venues:

- Conferences: FOCS '22; CCC '22; CCC '23; SAT '23; SAT '24; ICALP '24 (x2); AAAI '23; AAAI '24; SOSA '25; STOC '25
- Journals: ACM Transactions on Computational Logic; Theoretical Computer Science; SIAM Journal on Discrete Mathematics

Teaching Experience

Lund University

- EDAA40/75 Discrete Structures in Computer Science, 2022, 2024, 2025
- EDAN55 Advanced Algorithms, 2023, 2024
- EDAN01 Constraint Programming, 2022, 2023

KTH Royal Institute of Technology

- SF1661 Perspectives on Mathematics, 2018
- SF1624 Algebra and Geometry, 2017

Work Experience

AP3 Third National Swedish Pension Fund

ILS Intern

Quantitative analysis of insurance-linked securities (ILS).

Stockholm, Sweden

2019–2021

Technical Skills

Programming: Python, Go, $\text{\LaTeX}$, Julia, MATLAB

Software: RMS Miu